The Big Five Are Not Simple Composite Scores: Causal Complexity and Multilevel Trait Structure: A Commentary on Grosz (2026)

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Abstract

Grosz (2026) argues that the Big Five should be understood as composite-formative constructs rather than as latent common causes of their indicators. We agree that broad personality domains are unlikely to be single biological, environmental, or psychological entities that directly generate all item responses. Yet this critique of simple reflective models does not establish that the Big Five are merely composites. Statistical unidimensionality does not require mechanistic singularity. Broad dimensions emerge from numerous partly shared biological, environmental, cognitive, motivational, social, and behavioral influences. Indeed, multiplicity of causes is the reason why continuous, approximately Gaussian traits are plausible. Hierarchical latent models can represent this distributed structure while retaining substantive variance at the facet, nuance, item, method, and residual levels. Local dependence and semantic overlap challenge overly simplistic measurement models, not latent-variable thinking as such. Most importantly, the covariance among personality indicators remains to be explained. Composite scores preserve useful summaries and may predict outcomes well, but they do not explain why the summarized indicators hang together. Treating compositing as a solution therefore risks an operationalist retreat from understanding personality structure. We propose a multilevel causal-measurement framework in which broad domains capture approximate higher-order regularities, narrower tendencies retain their specificity, and the processes generating their covariance become explicit objects of inquiry.

Keywords: latent variables; composite scores; network analysis; Big Five; personality

The Big Five Are Not Simple Composite Scores: Causal Complexity and Multilevel Trait Structure: A Commentary on Grosz (2026)

Grosz (2026) offers an important critique of how the Big Five are often interpreted in personality psychology. The critique is useful because it makes explicit an assumption that is too often left implicit: that a broad personality factor such as extraversion, agreeableness, conscientiousness, neuroticism, or openness is a common cause of the items used to measure it. In a strict reflective model, the latent trait causes item responses, and associations among items are expected to disappear once the latent trait is controlled. Grosz argues that this picture is implausible for the Big Five. We agree. The Big Five are unlikely to correspond to five simple biological, environmental, or mental entities that directly produce responses to all their indicators.

Our disagreement concerns the stronger conclusion that follows in Grosz's article. From the fact that the Big Five are not simple common causes, Grosz concludes that they should instead be understood as composites of personality items and modeled, for descriptive and predictive purposes, with composite-formative models. This inference does not follow. The critique successfully challenges a naive version of reflective realism, but a composite-only conclusion requires an additional step, and that step is missing. In our opinion, a more plausible conclusion is that Big Five domains are higher-order dispositional structures: stable, reproducible patterns of covariation among narrower tendencies, generated by many partly shared biological, environmental, cognitive, motivational, social, and behavioral mechanisms. This reading is descriptively continuous with the original taxonomic definition of the domains as clusters of covarying tendencies (McCrae & John, 1992), and it adds the explanatory claim that the covariance has shared causes that a latent model can represent.

Latent Dimensions Need Not Be Single Mechanisms

Grosz's argument relies on a demanding interpretation of what a latent common cause must be. If a reflective model requires one and only one biological, environmental, or mental entity that causes all item responses, then the model is indeed implausible. Extraversion is not a single small element inside the person that pushes upward talkativeness, sociability, assertiveness, excitement seeking, and positive affect. On this point, Grosz is right.

Statistical unidimensionality and mechanistic singularity, however, are two different things (Borsboom et al., 2003, 2004; Eronen, 2025). A latent factor can be treated as a model-based representation of shared variance generated by many partly overlapping causes. These causes need not be exactly identical across all indicators. Nor must they operate through perfectly proportional pathways. What matters is whether the shared influences are strong and aligned enough to make a broad dimension useful for organizing individual differences. In this sense, a Big Five factor is a compact representation of a distributed causal structure. In a structural model, the factor stands in for a constellation of causes whose joint effects on the indicators are approximately proportional. Approximate proportionality is the relevant assumption; mechanistic singularity is unnecessary. Under this view, conscientiousness summarizes a distributed set of potentially separate and independent processes that help generate orderliness, diligence, self-discipline, and dutifulness.

A latent variable is especially useful when shared causes are distributed and not individually identifiable. When those causes can be isolated and measured, they can be modeled directly. Until then, the latent factor represents their common causal background. Showing that the causes behind a Big Five domain are numerous and distributed therefore describes the very situation in which latent modeling becomes useful. A list of separate predictors would be preferable only if the relevant causes had already been identified, measured, and theoretically separated.

Causal complexity is the expected case in personality psychology, and in individual differences more generally. Genetic studies do not suggest that Big Five domains map onto single genetic variants or isolated biological mechanisms. Rather, they point to highly polygenic and heterogeneous architectures. Behavior-genetic work similarly suggests a multilevel structure in which domain-general influences coexist with facet-specific influences. Grosz cites Franic, Borsboom, Dolan, and Boomsma (2014) for one half of their result: models allowing genetic and environmental factors to act directly on the items fit better than models in which a single latent variable per domain fully mediates those effects. The other half of the same study points the other way. Franic et al. also recovered five-factor genetic and environmental covariance structures. The shared influences on the items are themselves organized along the Big Five. Read in full, the study supports a multilevel structure in which a distributed domain-level background coexists with facet- and item-specific influences. Briley and Tucker-Drob (2012) reach the same multilevel conclusion from twin data. The evidence Grosz invokes against the latent domain therefore requires a higher-order representation, although a richer one than the single-mediator model he rejects.

The widespread assumption of normally-distributed latent factors makes this point particularly explicit. Under familiar central-limit conditions, a Gaussian distribution is the expected result of many small, partly independent influences combining additively. It is therefore theoretically at odds with the image of a latent factor as the expression of one singular mechanism. Factor analysis and item-response models have long treated broad latent differences as continuous precisely because they can arise from the accumulation of many genetic, environmental, neurobiological, and psychological causes. Grosz's evidence for causal multiplicity consequently weakens a simple single-mechanism interpretation, while fitting naturally with a distributed latent-trait account. Such complexity may require multilevel latent structures, networks of narrower tendencies, or combinations of these approaches.

The appropriate question is therefore quantitative and comparative. How much covariance is captured by the broad domain? How much remains at the facet, nuance, item, method, or residual level? Are deviations from proportionality large enough to undermine the use of a broad factor for a given purpose? Which model best captures the relevant structure for the question at hand? These questions cannot be answered by noting that the Big Five are not single entities. They require explicit multilevel modeling.

The condition of sufficiently aligned effects is also empirically constrained. Strong deviations should appear as poor fit, large residual correlations, cross-loadings, local factors, or unstable parameter estimates. These diagnostics are the everyday language of factor analysis and SEM, where departures from simple structure have long been expected, detected, and modeled. They may reflect non-proportional substantive causes, semantic redundancy, wording effects, response styles, or method variance. The appropriate response depends on the source: facet factors, method factors, residual covariances, revised item wording, or a narrower target construct may be appropriate (Leising et al., 2025; Soto & John, 2017). This keeps the distributed common-cause view open to failure for specific instruments and purposes.

Composite scores remain useful for description and prediction, but they leave the covariance to be explained. If two narrow tendencies covary and neither causes the other, their covariance points to shared causes (Reichenbach, 1956). A general latent factor can represent that shared background, while remaining only one candidate explanation. Leaving facet covariance as an unexplained correlation is therefore a theoretical retreat: the composite records the phenomenon without explaining it.

Separability and the Blood-Glucose Standard

A second main point of Grosz's critique partly rests on a separability argument. Blood glucose can be identified independently of the instruments used to measure it, and variations in blood glucose can plausibly be treated as common causes of readings obtained from different indicators. Big Five traits do not have this structure. Extraversion cannot be isolated from talkativeness, sociability, enthusiasm, assertiveness, activity, and related tendencies in the way blood glucose can be isolated from glucose meters. Grosz takes this as evidence that Big Five domains are composites and that latent-trait interpretations are unwarranted.

Blood glucose is itself multicausally determined, so causal simplicity cannot explain why it works better as a measurement benchmark. The blood-glucose analogy nevertheless remains poorly matched to any psychological construct. What distinguishes blood glucose is the precision with which a specific physiological quantity can be isolated and measured. A syndrome or disease would be a closer analogy, although still imperfect. Many clinical conditions have multiple causes and heterogeneous manifestations, yet they can remain meaningful units when their manifestations converge into a stable pattern. Personality traits are even more continuous and probabilistic, so a single-quantity benchmark is too strict. The benchmark also targets the wrong question.

In psychology the object of interest is rarely the observed behavior itself. It is what that behavior indicates, and the level at which the shared variance behind the indicators explains other outcomes. For general intelligence, this has long been familiar: vocabulary or matrix items matter because of what they jointly indicate and predict, not just because a person possesses a large vocabulary. Personality may sit closer to the observable end, and this should be argued on its own terms. Even so, the inferential target remains the shared level, not the isolated item.

Whether a Big Five domain is physically separable from its indicators is therefore secondary. The domain-level variance may carry explanatory and predictive weight that item-specific residuals do not. Narrow personality traits do not usually have this kind of measurement status. Talkativeness, sociability, assertiveness, or orderliness are inferred from self-reports, other-reports, and context-dependent behaviors, each of which contains domain-level variance, narrow substantive variance, method variance, response style, and random error. This asymmetry is decisive against the glucose analogy. Blood glucose is isolable because the specific quantity is measured with high precision. Narrow personality traits often lack this status. Once the broad domain is partialled out, the reliable trait-specific variance that remains in a single facet or item can be modest (Westfall & Yarkoni, 2016). Personality indicators are often measured reliably at the domain level, while the narrow level that Grosz wants to move to remains noisy. His own recommendation to estimate item retest reliabilities concedes this point: items carry substantial measurement error that only aggregation and explicit error modeling can address. After the broad domain is partialled out, the specific reliability of a narrow facet or item can be limited, even though nuance-level characteristics can show cross-rater agreement, rank-order stability, heritability, and practical utility when assessed and aggregated appropriately (Mõttus et al., 2019; Westfall & Yarkoni, 2016). Separability in personality is therefore graded and model-dependent. Broad domains remain useful because they stabilize and organize shared variance that narrow indicators often capture only noisily.

The related mereological claim is also underdetermined. One may say that narrow traits are “parts” of extraversion while being distinct from effects of extraversion. But part-whole language does not explain why the parts covary. If talkativeness, sociability, assertiveness, positive affect, and reward sensitivity form a stable pattern across persons, instruments, and occasions, this pattern remains an explanandum. It may reflect common causes, reciprocal reinforcement, semantic overlap, social learning, biological dispositions, or mixtures of these. A composite description does not identify which of these mechanisms is responsible. Separability arguments therefore challenge a simple entity view of the Big Five, but they do not by themselves justify a composite-only interpretation.

Local Dependence Requires Richer Measurement Models

Grosz also emphasizes that Big Five items are linked by direct causal relations, semantic overlap, and logical dependencies. This is a serious challenge to simple reflective models. If enthusiasm increases talkativeness, if being organized facilitates being punctual, or if two items are near paraphrases, then local independence is violated. A model in which all item associations are fully explained by one latent variable will be too simple and Grosz is, again, right.

Nonetheless, the conclusion should remain limited. The first question is inappropriateness of what: a specific simple CFA model, a particular item set, or the broader idea that shared variance can be modeled? The evidence usually speaks most directly to the first two (Rhemtulla et al., 2020). Local dependence leaves several explanatory representations open, including network models, multilevel latent structures, and combinations of these approaches. Composite scoring remains available for description and prediction, although it does not decompose the sources of local dependence. Direct item relations, semantic overlap, and residual dependencies can be modeled with facets, correlated residuals, method factors, specific factors, network edges, cross-loadings, or hierarchical structures. Local dependence therefore motivates richer models and stronger item-level diagnostics. The distinction between two sources of local dependence defuses the charge of post-hoc curve-fitting. If items covary beyond the factor because of semantic overlap or near-duplication, a correlated residual is a stopgap and the proper fix is to repair the instrument for the next administration. If they covary because a genuine facet sits between item and domain, the corresponding structure is hierarchical variance predicted in advance. A higher-order model that anticipates general plus specific variance is a specification, not an excuse.

Here, again, a methodological analogy with hierarchical models of intelligence is useful: specific subtest's variance never invalidated the abstractive and predictive value of the general factor because the hierarchy was expected from the start. The empirical question is the level at which external outcomes are explained.

Rhemtulla et al. (2020), cited by Grosz, show that imposing a reflective model on data not generated reflectively biases estimates. We accept the result and read it precisely: it indicts the wrong reflective model, the simple one-factor specification with local independence. It does not indict the hierarchical specification that expects the dependence. The remedy their result motivates is a better structural model, or a better causal/process model, not the abandonment of structure for a sum.

Leising et al. (2025) make this ambiguity especially clear. Correlations among person-descriptive judgments may reflect common substantive causes, direct substantive relations among target characteristics, semantic redundancy, perceiver attitudes, response styles, or mixtures of these. This multiplicity blocks simple inferences from item covariance to any single representation. Item covariance may support a common-cause model, a network model, a method model, or a multilevel combination of these explanatory representations (Cramer et al., 2012). Composite scores remain useful descriptive summaries of the same covariance. The covariance remains explanatorily ambiguous until its sources are decomposed.

The same reasoning applies to semantic overlap. Semantic similarity among items can inflate associations and make factor structures partly linguistic. Yet semantic overlap is not a reason to stop modeling latent structure. It is a reason to distinguish semantic methodological redundancy from substantive covariation, to use multimethod data when possible, and to test whether the same structure appears across item formats, informants, behaviors, and contexts. A composite score formed from semantically overlapping items does not remove this problem. It carries the redundancy into the score without further questioning it.

Composite-Formative Models as an Operationalist Retreat

Composite-formative models have legitimate uses. When the aim is to summarize a set of observed variables or maximize prediction of an outcome, a weighted or unweighted composite can be transparent and useful (Bollen & Bauldry, 2011). Grosz is right that composites avoid one strong assumption of reflective models: they do not require the indicators to be effects of a latent common cause.

This agnosticism should not be confused with explanation. The composite-formative model is most defensible as an agnostic scoring convention. It becomes less convincing when it is treated as the preferred ontological interpretation of the Big Five. A composite-formative model defines the Big Five score as an algebraic function of its indicators. This is useful as a scoring rule, although it does not explain why the indicators cohere. If items assessing talkativeness, sociability, enthusiasm, and assertiveness correlate robustly enough to justify an extraversion score, their covariance remains a substantive phenomenon. Composite scoring preserves that phenomenon in a convenient numerical form; it does not explain it. The reason is that a sum, taken on its own, leaves the covariance among the indicators unexplained. By Reichenbach's common cause principle, indicators that covary without causing one another share common causes. A reflective or hierarchical model names that shared background and models it. The same is done, yet differently, by process models. A composite absorbs it into a score and is silent about it. The composite is a legitimate scoring rule sitting on top of an unstated causal structure. The question is what that structure is, and a sum does not answer it.

Composite-formative models risk becoming an operationalist retreat when they are offered as a general replacement for latent-trait models (Yarkoni, 2020). Composites are often useful, especially for transparent scoring and prediction. A descriptive scoring rule can still be mistaken for a solution to an explanatory problem. Saying that extraversion is the sum of extraversion items tells us how the score is computed. It leaves open why these items covary, why the covariance is stable, why the score predicts life outcomes, why it shows genetic associations, and why it can be recovered across instruments and samples. Treating covariance as theoretically informative gives a rationale for grouping these indicators instead of arbitrary alternatives. The fallback position of “correlated but distinct facets”, sometimes offered as a theory-neutral middle ground, is the weakest option. A correlation is not an explanation. Two facets correlate because one causes the other, because they share causes, or because both processes occur together. Recording the correlation while leaving these possibilities unmodeled explains nothing and forfeits the ability to separate true score from measurement error. Stopping at correlated facets is the point at which explanation and understanding are abandoned. Similarly, simulating correlated variables without specifying a data generating process of such correlations is misleading and inappropriate.

A systems view can make a composite principled. If one holds that the indicators covary because of reciprocal causation, social learning, and partly shared architecture, then a composite may summarize a system whose dynamics are specified elsewhere. That position is coherent. The weaker move is adopting the composite while leaving the underlying model unstated, which keeps the descriptive convenience of a sum and the explanatory commitments of none.

Direct relations among indicators also complicate composite interpretation. Suppose enthusiasm increases talkativeness, talkativeness elicits rewarding social interaction, and rewarding social interaction further increases enthusiasm. A composite of enthusiasm and talkativeness can summarize this pattern, but it does not tell us whether its components are independent ingredients, consequences of one another, or products of a feedback loop. If one component partly causes another, a composite may weight the upstream process both directly and indirectly through its downstream consequences. With reciprocal causation, the score reflects an equilibrium of mutually reinforcing processes and does not have a simple additive interpretation.

This point does not make composites mathematically invalid. It does make their substantive interpretation less transparent. The same local dependencies that challenge simple reflective models also limit the explanatory interpretation of simple composite scores. They point toward causal and explicit modeling of the indicator system.

The Positive Manifold Cannot Be Merely Aggregated Away

A central fact about personality data remains: indicators within broad domains tend to covary. This positive manifold per se is not proof of a latent common cause (i.e., a set of shared causes), although that remains our favorite explanation. It may also arise from direct reciprocal relations, semantic overlap, social desirability, response styles, or mixtures of these. Yet the positive manifold is an explanandum. One further observation bears on the composite interpretation: if the Big Five were simple algebraic sums tied to specific item sets, their predictive and structural properties should change substantially when the item set changes. The evidence runs in the opposite direction. Domain-level convergence across instruments with different item pools, response formats, and informants is well documented, and the five-factor structure recovers across languages and cultures. This cross-instrument approximate invariance is difficult to explain if each composite is merely the sum of its own particular items; it is consistent with the view that different instruments sample, imperfectly and from different angles, the same underlying covariance space.

The existence of a useful composite presupposes that the indicators hang together; otherwise there would be little reason to aggregate them. A composite faces a simple dilemma. If its indicators do not covary, its psychological unity is unclear. If they do covary, that covariance requires explanation and cannot be made theoretically irrelevant by summing the indicators.

The composite-only interpretation therefore faces a limitation. It can describe the manifold, but it does not explain it. If the correlations among indicators are due mainly to common substantive causes, then some form of common-cause or shared-variance explanation re-enters. If they are due mainly to reciprocal relations among narrower tendencies, then a network or process model is needed. If both sources are important, latent and network models may need to be integrated. If they are due mainly to semantic overlap, then the issue concerns item construction and language. If they are due to perceiver’s attitudes or response styles, then method variance must be modeled. If they are due to how the instrument and the scales are constructed, then the measure is the problem. In none of these cases is aggregation by itself explanatory.

This point also preserves the distinction between prediction and explanation. A composite or item-level algorithm may predict an outcome well while telling us little about why the predictors covary, why they predict the outcome, or what construct the predictor actually represents. Conversely, a broad factor may organize a domain of individual differences even when narrower indicators outperform it for some specific outcomes. Prediction and explanation are related but not identical aims. Treating the Big Five as composites may be sufficient for some predictive purposes, but it does not settle their theoretical status.

A latent-variable model can fail when its causal assumptions are false. A composite-formative model can also fail when it treats a heterogeneous, locally dependent, or semantically redundant item set as a straightforward additive score. Repeated items, logical opposites, semantic overlap, and direct causal relations among indicators remain present under compositing and are absorbed into the score. Composite-formative models therefore leave important assumptions implicit. The relevant question concerns which model best represents the covariance structure, residual dependencies, and substantive theory of the construct. A composite-formative modeling is not a safe, assumptions-free solution.

The Intelligence Analogy Clarifies the Composite-Only Inference

The same logic appears in intelligence research. Cognitive tests show a positive manifold, task-specific variance, group factors, method effects, training effects, and local dependencies. General intelligence is also genetically and biologically distributed, with many small influences and no single simple mechanism (Plomin & Deary, 2015; Savage et al., 2018; Spearman, 1904). These facts have not led intelligence research to treat general intelligence as an arbitrary sum of test scores.

The usual response has been hierarchical and process-oriented modeling. The general factor, broad abilities, narrow abilities, and task-specific residuals can be represented together in hierarchical models (Carroll, 1993). Alternative accounts, such as mutualism, also treat the positive manifold as a phenomenon requiring explanation (Van Der Maas et al., 2006). The shared lesson is that covariance structure remains theoretically informative even when its generative mechanisms are distributed and complex.

The intervention literature sharpens the same methodological point. Working memory training can improve the trained task while showing little transfer to general cognitive ability (Melby-Lervåg et al., 2016). This cuts against a hasty composite reading. The specific skill could be moved, yet because the movement did not propagate to the general factor, the intervention lost much of its theoretical interest. If g is the level that carries consequential variance and training cannot shift g, then shifting the narrow component is a local effect of limited reach. Transposed to personality, and only as a methodological analogy, an intervention that changes one facet without moving the domain shows that the manipulation was local. Whether that local change matters depends on whether it propagates to the level at which outcomes are explained. This supports joint modeling of the levels.

Grosz himself judges mutualism an unlikely primary source of within-domain covariance in personality and points instead to shared genetic and biological influences, including Franic et al. (2014) and Jackson and Wright (2024). We take this seriously, and it forces a dilemma that Grosz does not resolve. We do not adopt mutualism for personality; we raise it as an explicit benchmark to make the choice visible. Either the covariance among Big Five indicators is generated by a declared systems model, with reciprocal causation in the mutualist sense, in which case the generative theory is a network and its implications must be accepted in full. Or the covariance is generated mainly by shared causes, which is the option Grosz's own cited evidence favors, in which case a distributed common cause returns to the center of the explanation and a higher-order latent factor is its natural representation. The third path is the weak one: adopting an agnostic composite while invoking shared genetic and biological causes informally, taking the explanatory credit of common causes without paying the modeling cost of representing them. A composite is an operationalist retreat precisely when it occupies that third path.

Causal Inference Is Multilevel

Grosz is right that broad personality domains are often poor direct intervention targets. On the narrow point that the Big Five are weak treatment variables, we do not contest him. We make a different and complementary claim: ruling the domains out as treatments does not rule them out of causal models in every role. Conflating "a poor cause to manipulate" with "a variable to delete from the model" is the step we resist. An intervention rarely changes extraversion as a whole. It changes opportunities, habits, skills, self-beliefs, motives, or specific behaviors. Narrow traits and multimethod assessments can therefore be closer to actionable mechanisms. This recommendation is fully compatible with latent-variable thinking. Self-reports, informant reports, and behavioral recordings can be treated as fallible indicators of a narrower disposition, with method variance modeled explicitly.

Broad domains can still matter for causal inference in other roles. A broad domain may represent shared etiological variance that confounds the relation between a narrow trait and an outcome. It may moderate the effect of an intervention on a facet. It may serve as a mediator, a higher-order outcome, or a background liability that shapes many downstream behaviors at once. VanderWeele's (2022) discussion of constructed measures is useful here because causal inference often concerns multivariate constructs whose components have different causal relations with outcomes. Westfall and Yarkoni (2016) make a related point for statistical control: when constructs overlap and are measured fallibly, naive adjustment can seriously distort inferences. These possibilities require explicit causal diagrams and measurement models. Removing broad traits from the model can obscure the source of covariance that links the narrow variables together. A related concern in this literature is collider bias: Wysocki et al. (2022) argue that statistical control of a composite-formative score can introduce selection distortions because the components of an arbitrary composite may not share a common causal pathway. This concern, however, is not symmetrically a problem for the distributed common-cause view we defend. If a broad domain represents a higher-order latent structure generated by many partly shared causes, then conditioning on that domain in a causal model amounts to conditioning on the shared causal background of the facets, not on an arbitrary aggregate. The relevant question is therefore whether the broad trait being controlled for is a genuine common-cause representation or an atheoretical sum. For a distributed latent factor, controlling for the domain-level variance is the appropriate move to close backdoor paths through the shared etiology; for a purely composite score with no declared causal structure, the concern raised by Wysocki et al. applies. This distinction motivates explicit modeling of the higher-order structure rather than its removal.

A reviewer might object that conditioning on a broad factor risks collider or M-bias. The objection changes once the cases are separated. If the facets are uncorrelated, no one would introduce a common factor over them, and the worry is moot. If the facets correlate because of a shared distributed background, then that background is a common cause of the facets and a confounder of the facet-to-outcome relation whenever it also affects the outcome. Representing it, through a higher-order factor that stands in for the shared causes, is what closes the backdoor path. This does not make adjustment automatic. It means that the bias objection is not asymmetrically a problem for hierarchical latent models. The right adjustment depends on the causal graph, which depends on the theory.

For example, suppose talkativeness predicts occupational networking. If this association is estimated without modelling broader extraversion-related variance, the result may conflate talkativeness-specific effects with the influence of broader tendencies such as sociability, reward sensitivity, positive affect, or social confidence. If a broad extraversion factor is theoretically assumed, it should therefore be represented in the model. The factor here is not posited as an entity. It is the model-based representation of the shared causal background that makes the facets covary in the first place, the same object defended in the opening sections. Controlling for it means controlling for that shared background, not for a free-floating statistical abstraction. Doing so does not make the facet irrelevant; rather, it decomposes the association into a domain-general component and a facet-specific residual component. The key question is then whether occupational networking is predicted by broad extraversion, by the talkativeness-specific residual, or by both. Causal inference in personality is therefore multilevel: items, facets, domains, methods, and mechanisms may each be relevant depending on the estimand.

This is also why facet-level heterogeneity should not be interpreted too quickly. If facets within a domain predict an outcome in different directions, this may indicate that only narrow traits matter. But it may also indicate multiple indirect pathways from a broader disposition through different components. Extraversion, for instance, could increase social contact through sociability while also increasing interpersonal conflict through dominance. Opposing indirect effects would make the total domain-level effect small, even if broader dispositional structure remains causally relevant. Facet-level analysis is therefore essential, but it does not eliminate the need to model higher-order structure.

Toward Multilevel Causal-Measurement Models

The constructive alternative is concrete and already implementable: hierarchical structural equation models that separate measurement error from true variance, represent the distributed domain-level background at the top, and preserve facet- and nuance-level specificity below. This is a standard hierarchical SEM toolkit, applied with explicit ontological assumptions. Method factors, correlated residuals for semantic redundancy, and direct dependencies enter where theory and item design predict them, with post-hoc repairs reserved for cases of misfit. The Big Five can still organize a wide trait space, including many narrower trait scales, while leaving room for intermediate-level aspects, facets, and nuances (Bainbridge et al., 2022; DeYoung et al., 2007; Soto & John, 2017).

Whole Trait Theory provides a useful bridge between descriptive trait structure and process explanation by distinguishing stable distributions of personality states from the social-cognitive, motivational, affective, biological, and situational processes that generate them (Fleeson & Jayawickreme, 2015). This distinction is compatible with hierarchical measurement: broad domains can represent stable between-person regularities without being treated as unitary mechanisms that directly cause momentary states. On this interpretation, latent trait dimensions belong primarily to the descriptive level, whereas their covariance and stability require explanation through lower-level processes (Wood et al., 2015). Whole Trait Theory therefore illustrates why rejecting a simple common-cause interpretation need not entail abandoning broad latent structure, although it does not by itself determine the appropriate measurement model.

This position is compatible with the most important lesson of Grosz's critique. The Big

Five should not be treated as five simple entities analogous to blood glucose or temperature. Their indicators are impure reflections of many partly shared and partly specific processes. Local dependence, semantic overlap, facet-specific variance, and causal heterogeneity are real. These facts support a pluralistic, explicit, and multilevel account of broad traits as model-based representations of a distributed causal background.

Conclusion

Grosz's critique is valuable because it challenges an overly simple interpretation of Big Five measurement. Personality psychologists should not assume that a broad factor is a single common cause of its indicators, nor should they ignore local dependence, semantic overlap, or facet-level heterogeneity. These points are important.

The problem is that the conclusion that the Big Five should be treated as composites is underdetermined. Statistical unidimensionality does not entail mechanistic singularity. A latent dimension can summarize a distributed causal structure generated by many partly shared mechanisms. Local dependence challenges simple models. Latent structure remains a live explanatory hypothesis. Composite-formative scores are useful descriptive tools, although they do not explain the covariance structure that makes them useful. The positive manifold cannot be aggregated away.

It cannot, because covariance among indicators that are not causally related to one another implies shared causes, and shared causes are what a hierarchical latent model represents and a sum conceals. The choice Grosz leaves open should be closed honestly: either a declared systems model behind the composite, with its full implications, or a distributed common cause and the higher-order structure that represents it. The agnostic composite, invoking shared causes while modeling none, is the one position the evidence does not support.

The more defensible conclusion is that causal complexity challenges simple reflective models. Latent traits remain useful when they are treated as higher-order, multilevel, and approximate representations. Big Five domains are broad, imperfect, and causally heterogeneous structures that organize individual differences and guide multilevel explanation. The task is to model explicitly the levels at which personality structure, prediction, and causal explanation each operate.

Declarations

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